Troubleshooting Guide: Amine Regenerator — Reboiler Fouling and Stripper Foaming Carryover

Category: Gas Treating / Chemical Regeneration **Unit:** Amine Regenerator (Stripper) and Reboiler **Tools:** HYSYS/UniSim Amine Package regenerator model, reboiler duty/U-value trending, amine degradation product analysis **Fluid Package:** Amine/Acid Gas package (reactive absorption chemistry) — the same specialized package family used for the absorber (Case Study 9), needed because regeneration also involves amine-CO_2/H_2S chemical equilibrium, not simple physical stripping **Symptom:** Lean amine concentration gradually declining despite stable reboiler steam/duty input, with occasional foam observed in the stripper overhead

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1. Quick Reference

Item	Detail
Reported symptom	Lean amine concentration gradually dropping despite stable reboiler duty input; occasional stripper overhead foaming observed
Initially unclear	Whether this was a reboiler performance issue (fouling reducing effective duty) or a stripper hydraulic/foaming issue reducing stripping efficiency
Actual root cause	Reboiler tube fouling (from amine degradation products/heat-stable salts) reduced the effective heat transfer coefficient, lowering actual stripping steam generation even though duty setpoint/steam flow appeared normal — and the resulting poor stripping performance increased degradation products further, which also promoted stripper foaming
Fix	Mechanically cleaned reboiler tubes; implemented amine reclaiming to remove heat-stable salts and degradation products
Diagnostic signal	Reboiler duty/steam flow looked normal, but the calculated overall U-value (from actual temperatures and heat duty) had dropped well below clean design value
Prevention	Routine reboiler U-value trending; periodic heat-stable salt (HSS) analysis with a reclaiming program tied to HSS concentration

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2. Symptom

- **Lean amine concentration gradually declined** over time, even though **reboiler steam/duty input appeared stable** at its normal setpoint.

- **Occasional foam was observed in the stripper overhead.**

3. Why This Wasn't Assumed to Be a Simple Reboiler Duty Shortfall

A declining lean amine concentration with "stable" reboiler duty seems contradictory at first — if duty is stable, why would stripping performance decline? This apparent contradiction was the key clue: **steam flow/duty setpoint being stable doesn't guarantee the same amount of heat is actually being transferred to the amine**, if the reboiler's heat transfer surface itself has degraded. This needed the same "duty delivered vs. duty commanded" distinction seen in the distillation column case (Case Study 4), but for a reboiler fouling mechanism rather than an electrical fault.

4. Diagnostic Approach

Step 1 — Confirm reboiler duty/steam flow at the utility side

Reboiler steam flow and pressure were reviewed and found to be at their normal, stable values — the utility side appeared to be delivering as commanded.

Step 2 — Calculate the actual overall heat transfer coefficient (U)

Rather than relying on steam flow alone, the reboiler's **actual overall U-value** was calculated from measured amine-side and steam-side temperatures and the resulting heat duty, then compared to the **clean/design U-value** — the same core technique used in the seawater cooler case (Case Study 6).

Finding: Actual U had dropped **well below the clean design value**, meaning the reboiler was transferring significantly less heat to the amine than its steam consumption alone would suggest — a fouling signature, not a utility supply issue.

Step 3 — Investigate the fouling mechanism

With fouling confirmed via the U-value gap, the reboiler tubes were inspected and amine samples analyzed for **heat-stable salts (HSS) and degradation products**, both known amine reboiler foulants.

Finding: Elevated HSS/degradation product levels were confirmed, consistent with fouling deposits found on inspection.

Step 4 — Connect reduced stripping performance to the observed foaming

With reboiler heat transfer reduced, less stripping steam was actually being generated per unit of amine circulated, reducing acid gas stripping efficiency — which in turn tends to leave more degradation products in circulation (since they aren't being effectively purged), **compounding** the fouling over time and also contributing to the stripper foaming observed, since degradation products and heat-stable salts are well-known foam stabilizers (as in the absorber foaming case, Case Study 9, but here originating from regeneration-side chemistry rather than upstream carryover).

Quantitative Basis

- Reboiler steam flow held steady at **8,200 lb/hr, 50 psig** throughout — utility side delivering as commanded.
- Calculated actual U-value dropped from a clean design **165 Btu/hr·ft^2·°F to 98 Btu/hr·ft^2·°F — a 41% loss** — despite the unchanged steam supply.

- Lean amine concentration declined from a design **50.0 wt% MDEA to 46.5 wt%** over the same 4-month period.
- HSS analysis: design/action threshold is 2.0 wt% of total amine inventory; measured concentration reached **4.8 wt%.**

5. Root Cause

Reboiler tube fouling — driven by amine degradation products and heat-stable salts — reduced the reboiler's actual heat transfer coefficient, so less stripping steam was generated per unit of circulated amine even though commanded steam flow/duty appeared normal. This reduced stripping efficiency, which allowed degradation products to accumulate further (a compounding effect) and also promoted the observed stripper foaming, together driving down lean amine concentration.

6. Corrective Action

1. **Mechanically cleaned the reboiler tubes** to remove fouling deposits and restore heat transfer performance.
2. **Implemented amine reclaiming** to remove heat-stable salts and degradation products from the circulating amine inventory.

7. Verification

- Reboiler U-value returned to **158 Btu/hr·ft²·°F**, within 5% of clean design, following mechanical cleaning.
- Lean amine concentration recovered to **49.6 wt%**, and HSS dropped to **1.6 wt%**, below the 2.0 wt% action threshold, after the reclaiming program.
- Stripper foaming incidents dropped from **3 events/week to zero over the following 45 days.**

8. Prevention / Long-Term Fix

- Established **routine reboiler U-value trending** (not just steam flow monitoring) as the standard performance check, since steam flow alone can look normal while actual heat transfer degrades.
- Implemented **periodic heat-stable salt analysis with a reclaiming program tied to HSS concentration**, addressing the chemistry driving both the fouling and the foaming rather than treating them as separate issues.

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9. General Troubleshooting Checklist (Reusable)

- ☐ Don't treat stable steam flow/duty setpoint as proof the reboiler is delivering its intended heat — calculate actual U-value from measured temperatures and compare to clean/design value
- ☐ If U has dropped, inspect reboiler tubes and sample the amine for heat-stable salts and degradation products
- ☐ Recognize that reduced reboiler performance and stripper foaming are often linked through the same underlying chemistry (HSS/degradation products), not independent problems
- ☐ Mechanically clean fouled reboiler tubes
- ☐ Implement or increase amine reclaiming to remove HSS/degradation products from circulation

- ☐ Verify recovery via BOTH U-value trend AND lean amine concentration — a partial recovery in one without the other suggests the fix is incomplete
- ☐ Establish routine reboiler U-value trending and periodic HSS analysis as standing checks, since this failure mode compounds over time if left unaddressed

10. Key Takeaway

A reboiler delivering its commanded steam flow isn't necessarily delivering its intended heat duty — fouling can quietly reduce actual heat transfer while the utility-side numbers look completely normal. In amine systems specifically, reboiler fouling and stripper foaming are often two symptoms of the same underlying chemistry problem (heat-stable salts and degradation products), and reduced stripping performance from fouling can make that chemistry worse over time. Calculate actual U-value, don't just watch steam flow, and treat reclaiming as the fix for the shared root cause rather than treating fouling and foaming as separate issues.

Related Concepts / Tags

`amine-regenerator` `stripper` `reboiler-fouling` `heat-stable-salts` `HSS` `amine-degradation` `foaming` `U-value-trending` `amine-reclaiming` `gas-treating`

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.